Determination of the fine-structure constant with an accuracy of 81 parts per trillion

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The standard model of particle physics is remarkably successful because it is consistent with (almost) all experimental results. However, it fails to explain dark matter, dark energy and the imbalance between matter and antimatter in the Universe. Because discrepancies between standard-model predictions and experimental observations may provide evidence of new physics, an accurate evaluation of these predictions requires highly precise values of the fundamental physical constants. Among them, the fine-structure constant α is of particular importance because it sets the strength of the electromagnetic interaction between light and charged elementary particles, such as the electron and the muon. Here we use matter-wave interferometry to measure the recoil velocity of a rubidium atom that absorbs a photon, and determine the fine-structure constant $\alpha^{-1} = 137.035999206(11)$ with a relative accuracy of 81 parts per trillion. The accuracy of eleven digits in α leads to an electron g factor[1,2]—the most precise prediction of the standard model—that has a greatly reduced uncertainty. Our value of the fine-structure constant differs by more than 5 standard deviations from the best available result from caesium recoil measurements[3]. Our result modifies the constraints on possible candidate dark-matter particles proposed to explain the anomalous decays of excited states of ^{8}Be nuclei[4] and paves the way for testing the discrepancy observed in the magnetic moment anomaly of the muon[5] in the electron sector[6].

The fine-structure constant α is the pillar of our system of fundamental constants. As the measure of the strength of the electromagnetic interaction in the low-energy limit, it has been measured using diverse physical phenomena: the quantum Hall effect, the Josephson effect, the atomic fine structure, atomic recoils and the electron magnetic moment anomaly[7]. Comparison of results across sub-fields of physics is a powerful test of the consistency between theory and experiment. In particular, the fine-structure constant is a crucial parameter for testing quantum electrodynamics (QED) and the standard model. This test relies on the comparison between the measured value of the electron gyromagnetic anomaly $a_e = (g_e - 2)/2$ (where g_e is the electron g factor) and its theoretical value. The standard-model prediction $a_{e,SM}$ is dominated by the QED term given by a perturbation series of α/π, and contains additional contributions from hadronic and weak interactions. Numerical and analytical evaluations of the coefficients of the QED series are firmly established up to the eighth order, and the accuracy of the tenth order has been improved over the past years[1,2,8]. Assuming that the prediction of the standard model is correct, comparison of the theory with the most accurate measurement of the electron magnetic moment[9] leads to a value of the fine-structure constant with a relative accuracy of 2.4×10^{-10} dominated by experimental precision[9] (see Fig. 1).

From a different point of view, to test the prediction of the standard model, we need independent measurements of α with a similar precision to evaluate $a_{e,SM}$. The most successful independent approach is based on the measurement of the recoil velocity ($v_r = \hbar k/m$) of an atom of mass m that absorbs a photon of momentum $\hbar k$ (refs. [10,11]). Here $\hbar$ is the reduced Planck constant ($\hbar = h/(2\pi)$) and $k = 2\pi/\lambda$ is the photon wave vector, where λ is the laser wavelength. Such a measurement yields the ratio h/m and then α via the relation

$$\alpha^2 = \frac{2R_\infty}{c} \times \frac{m}{m_e} \times \frac{h}{m}. \quad (1)$$

The Rydberg constant R_∞ is determined from hydrogen spectroscopy with an accuracy of 1.9 parts per trillion (ppt; https://pml.nist.gov/cuu/Constants/). The atom-to-electron mass ratio m/m_e is obtained from the ratio of the relative atomic mass of the atom $A_r(m)$ (known at 69 ppt for rubidium[12,13]) and the relative atomic mass of the electron $A_r(m_e)$ (known at 30 ppt)[14]. The speed of light in vacuum, c, has a fixed value.

Here, we present a measurement of the recoil velocity on rubidium atoms. We measured $h/m(^{87}\mathrm{Rb}) = 4.59135925890(65) \times 10^{-9}\ \mathrm{m^2\ s^{-1}}$. In the international system of units adopted in 2019, in which h has a fixed value, we obtain $m(^{87}\mathrm{Rb}) = 1.44316089776(21) \times 10^{-25}$ kg. This is the most accurate atomic mass measurement so far, to our knowledge. This results leads to a fine-structure constant α of

$$\alpha^{-1} = 137.035999206(11).$$

The uncertainty contribution from the ratio $h/m(^{87}\mathrm{Rb})$ is 2.4×10^{-11} (statistical) and 6.8×10^{-11} (systematic). Our result improves the

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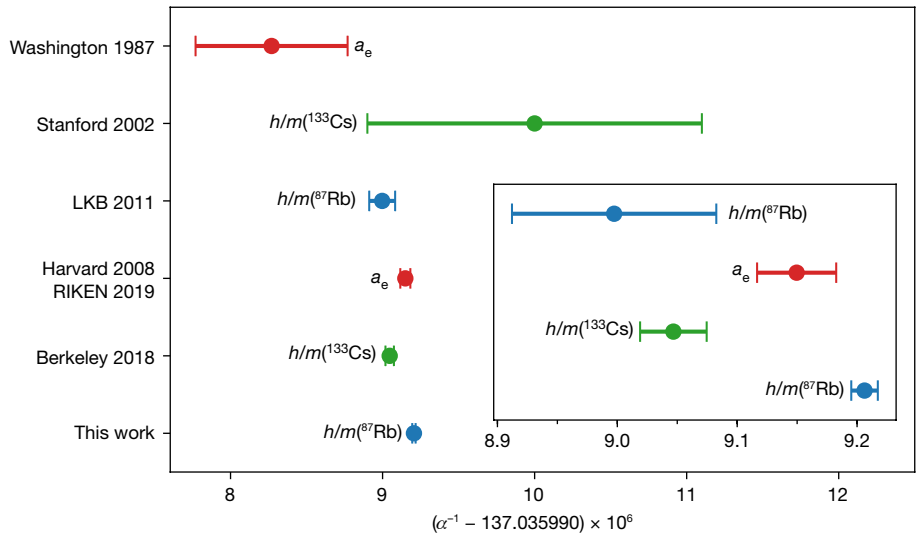


Fig. 1 | Precision measurements of the fine-structure constant. Comparison of most precise determinations of the fine-structure constant so far. The red points are from $g_e - 2$ measurements and QED calculations, and the green and blue points are obtained from measurements of caesium and rubidium atomic recoils, respectively. Errors bars correspond to $\pm 1\sigma$ uncertainty. Previous data are from ref. [34] (Washington 1987), ref. [10] (Stanford 2002), ref. [18] (LKB 2011), ref. [9] (Harvard 2008), ref. [2] (RIKEN 2019) and ref. [3] (Berkeley 2018). Inset, magnification of the most accurate values of the fine-structure constant.

accuracy on α by a factor of 2.5 over the previous caesium recoil measurement[3] but, most notably, it reveals a 5.4σ difference from this latest measurement.

We built a dedicated experimental setup and implemented robust methods to control systematic effects. By accelerating atoms up to 6 m s^{-1} in 6 ms and using typical two-photon Raman transitions as beam splitters for the matter waves, we obtained a relative sensitivity on the recoil velocity of 0.6 ppb in 1 h of integration (0.3 ppb on α). This sensitivity is more than three times better than that obtained using the best atom interferometer based on multi-photon beam splitters[3], although the latter technique is expected to provide a substantial gain in sensitivity with respect to Raman transitions[15,16].

The unprecedented sensitivity of our atom interferometer enables us to experimentally evaluate and mitigate several systematic biases. We recorded data with different experimental parameters, reinforcing the overall confidence of our error budget. We also implemented a Monte Carlo simulation that includes both the Ramsey–Bordé atom interferometer and the Bloch oscillations process. This code models precisely the underlying physics of our interferometer and provides an accurate evaluation of systematic effects, consistent with experimental results.

Experiment

Our experimental method is illustrated in Fig. 2. The basic tools of our experiment are Bloch oscillations in an accelerated optical lattice, which enable the coherent transfer of a precise number of photon momenta to the atoms (typically 1,000$\hbar k$), and a matter-wave interferometer that measures the phase shift due to the change in velocity of the atoms. As in the optical domain, atom interferometry needs tools to split and recombine atomic wave packets; this is accomplished by a sequence of light pulses. The probability of detecting atoms in a given internal state at the output of the interferometer is a sinusoidal function of the accumulated phase difference along the two paths. Thus, the measurement of atomic populations enables the evaluation of the phase shift. Using the combination of the Ramsey–Bordé interferometer configuration and Bloch oscillations, the phase shift is proportional to the ratio h/m (ref. [17]).

We produce a cold rubidium sample using an optical molasses in the main chamber. Then, atoms are transported to the interferometry area, a 70-cm-long tube surrounded by a two-layer magnetic shield. The magnetic field is controlled to within 50 nT. To that end, we use an atomic elevator based on two Bloch oscillation pulses (acceleration/deceleration)[17]. These are performed using two vertical counter-propagating laser beams, the frequency difference of which is swept to create an accelerated standing wave. Atomic trajectories are precisely adjusted by controlling this frequency difference. Between the two Bloch oscillation pulses of the elevator, we apply two Raman pulses to prepare atoms in a well defined atomic internal state (see Fig. 2b). Raman transitions occur between the two hyperfine levels of the ground state of the rubidium atom and are also implemented using two vertical counter-propagating laser beams (with wave vectors $\mathbf{k}_1 = -\mathbf{k}_2$ and $k_R = k_1 \approx k_2$). Their frequency difference ω_R is controlled to compensate precisely the Doppler shift induced by the accelerations of the atoms.

The atom interferometer is illustrated in Fig. 2c. It is implemented with two pairs of $\pi/2$ Raman pulses. Each pulse acts as a beam splitter by transferring a momentum of $2\hbar k_R$ to an atom with a probability of 50%. The first pair creates a coherent superposition of two spatially separated wave packets in the same internal state with the same momentum. The second pair recombines the two wave packets. Between the second and third $\pi/2$ pulses, a Bloch oscillation pulse transfers a momentum of $2N_B\hbar k_B$ to both wave packets, where N_B is the number of Bloch oscillations. The overall phase Φ of the interferometer is given by

$$\Phi = T_R\left[\varepsilon_R 2k_R\left(\varepsilon_B\frac{2N_B\hbar k_B}{m} - gT\right) - \delta\omega_R\right] + \phi_{LS}, \qquad (2)$$

where T_R is the time between the $\pi/2$ pulses of each pair, T is the time between the first and the third $\pi/2$ pulses, g is the gravitational acceleration, ϕ_{LS} represents the phase corresponding to parasitic atomic level shifts and $\delta\omega_R$ is the difference of the Raman frequencies between the first and the third $\pi/2$ pulses. ε_R and ε_B determine the orientation of Raman and Bloch lasers wave vectors, respectively.

The fluorescence signal collected in the detection zone gives the number of atoms in each atomic level at the output of the interferometer. Atomic fringes are obtained by measuring the fraction of atoms in a given internal state for varying $\delta\omega_R$. Using a mean-square adjustment, we calculate $\delta\omega_{R,0}$, the frequency for which $\Phi = 0$. Gravity is cancelled between upward ($\varepsilon_B = 1$) and downward ($\varepsilon_B = -1$) acceleration (see Fig. 2). Constant level shifts ϕ_{LS} are mitigated by inverting the direction of the

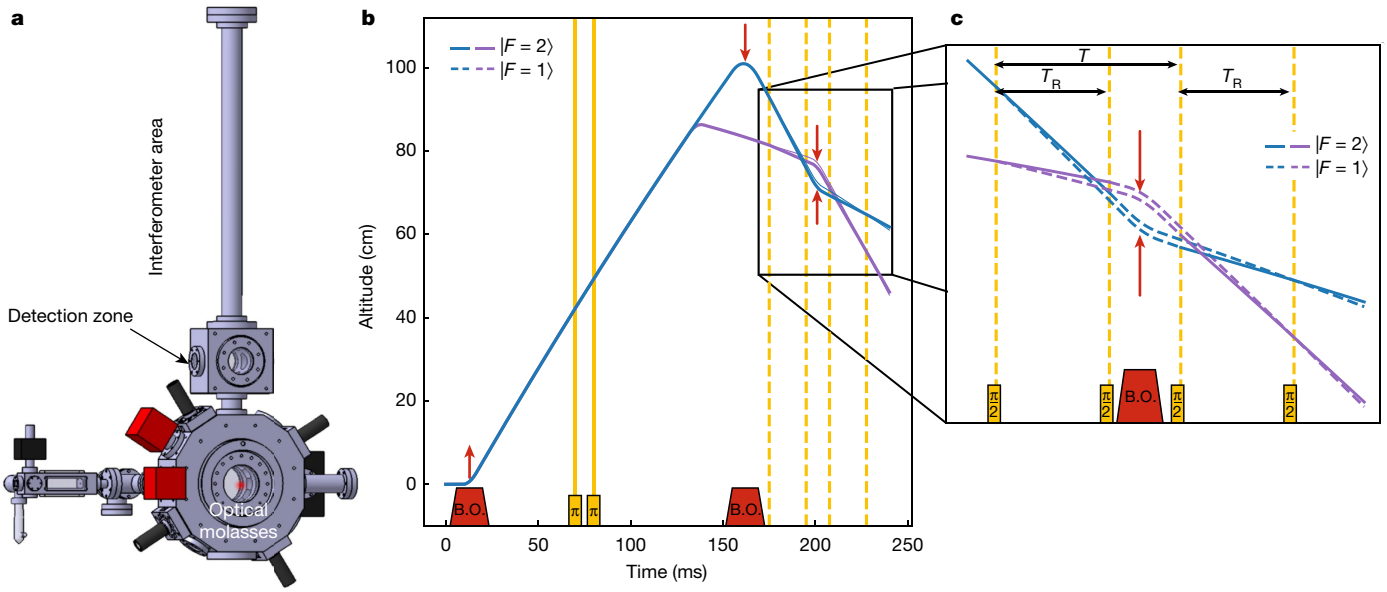


Fig. 2 | Experimental setup. a, Design of the vacuum chamber; the atom interferometer—a 70-cm-long magnetically shielded tube—is located in the upper area. **b**, Sequence of Bloch oscillations (B.O., red) and Raman pulses (yellow) used to control the trajectory of atoms before starting the atom interferometer. **c**, Atom interferometer light pulse sequence. The atomic trajectories for upward (blue) and downward (purple) accelerations are previously calculated to mitigate the gravity gradient effect. The separation between the two paths of each interferometer is exaggerated for clarity.

Raman beams ($\varepsilon_R = \pm 1$). The shot-to-shot parameters of the interferometer ($\delta\omega_R, \varepsilon_R, \varepsilon_B$) are applied randomly to avoid drifts. We record four spectra (Fig. 3a) that yield

$$\frac{\hbar}{m} = \frac{1}{4}\frac{\sum_{\varepsilon_R,\varepsilon_B}\left|\delta\omega_{R,0}(\varepsilon_R,\varepsilon_B)\right|}{4N_B k_B k_R}. \quad (3)$$

Data analysis

For the conditions of Fig. 3a, the typical uncertainty on $\delta\omega_{R,0}$ is 55 mHz. This leads to a statistical uncertainty on h/m of less than 2 ppb in 5 min. The behaviour of the Allan deviation calculated with a set of h/m measurements over 56 h (Fig. 3b) shows that the data are independent (no correlations or long-term drift). It also indicates that the sensitivity of our setup on α is 8×10^{-11} in 14 h.

Table 1 presents our error budget. Several systematic effects identified in our previous measurement[18] have been reduced by at least one order of magnitude. By controlling the experimental parameters of the atomic elevator, we are able to adjust precisely the altitude of atomic trajectories within 100 µm in such way that the gravity gradient cancels out between the configurations $\varepsilon_B = 1$ and $\varepsilon_B = -1$ (see Fig. 2c). The effect of Earth's rotation is suppressed by continuously rotating one of the Raman beams during the interferometric pulse sequence[19]. The long-term drift of the beam alignment is corrected with an accuracy better than 4 µrad every 45 min by controlling the retro-reflection of the laser beams via a single-mode optical fibre. Our lasers are locked on a stabilized Fabry–Pérot cavity and their frequencies are regularly measured using a frequency comb with an accuracy of less than 4 kHz. The low density of our atomic sample implies a reduction of the effects of the refraction index and atom–atom interaction[20] to less than 1 ppt. Effects related to the geometrical parameters of the laser beams (Gouy phase and wave front curvature) are mitigated by using a 4.9-mm-waist beam passing through an apodizing filter and by adjusting the curvature with a shearing interferometer.

Among the recently identified systematic effects, the most subtle one is related to correlations between the efficiency of the Bloch oscillations and short-scale spatial fluctuations in laser intensity. This effect raises the question of how to calculate the photon momentum in a distorted optical field. Relying on our previous work[21], we reduce the contribution of this effect to the error budget to less than 0.02 ppb. Because of the expansion of the atomic cloud, there is a residual phase shift that is due to the variation of the intensity perceived by the atoms. This phase shift depends on the velocity distribution[22,23]. We implement a method to compensate for the mean intensity variation and use a Monte Carlo simulation to evaluate the residual bias due to this Raman phase shift.

During the interferometer sequence, we apply a frequency ramp to compensate the Doppler shift induced by gravity. Nonlinearity in the delay of the optical phase-lock loop induces a residual phase shift that is measured and corrected for each spectrum. These systematic effects were not considered in our previous measurement[18] (see Fig. 1), which could explain the 2.4σ discrepancy between that measurement and the present one. Unfortunately, we do not have available data to evaluate retrospectively the contributions of the phase shift in the Raman phase-lock loop and of short-scale fluctuations in the laser intensity to the 2011 measurement. Thus, we cannot firmly state that these two effects are the cause of the 2.4σ discrepancy between our two measurements.

Overall systematic errors contribute an uncertainty of 6.8×10^{-11}. Figure 3c shows the data used for the determination of α. Each point represents about 10 h of data. We took advantage of the sensitivity and reproducibility of our setup to study systematic effects by varying the experimental parameters (such as pulse-separation time, number of Bloch oscillations, duration of Bloch pulse, laser intensity and atomic trajectories). In parallel, we performed theoretical modelling and numerical simulations to interpret the experimental observations. The measurement campaign lasted one year and ended when consistent values were obtained for the different configurations.

Using our measurement of the fine-structure constant, the standard-model prediction of the anomalous magnetic moment of the electron becomes

$$a_e(\alpha_{LKB2020}) = \frac{g_e - 2}{2} = 1{,}159{,}652{,}180.252\,(95) \times 10^{-12}.$$

The relative uncertainty on g_e is below 0.1 ppt, which is the most accurate prediction of the standard model. Comparison with the direct

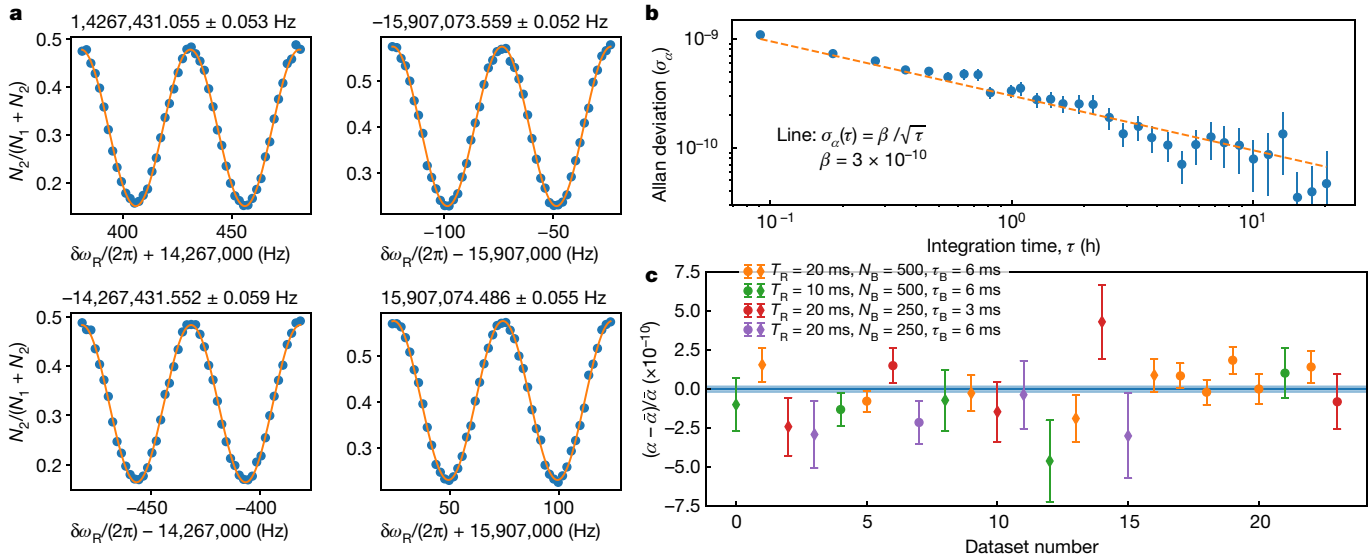


Fig. 3 | Data analysis. a, Typical set of four spectra recorded by inverting the directions of the Raman and Bloch beams for $T_R = 20$ ms and $N_B = 500$. Each spectrum displays the variation of the relative atomic population with respect to the parameter $\delta\omega_R$. The lines are least-squares fits used to determine the position of the central fringe displayed on the top of each spectrum. **b**, Allan deviation σ_α of the measurement of the fine-structure constant α at maximum sensitivity ($T_R = 20$ ms, $N_B = 500$) as a function of the integration time τ. The line corresponds to $\sigma_\alpha(\tau) = 3 \times 10^{-10}/\sqrt{\tau}$, with τ expressed in hours. Error bars indicate 1σ uncertainties. **c**, Datasets used to determine the value of the fine-structure constant, α. Data are obtained by changing the following experimental parameters: the pulse separation time, T_R, the number of Bloch oscillations, N_B, and their total duration, τ_B. The circles and diamonds correspond to two different laser intensities during the π/2 pulses of the interferometer. Error bars denote ±1σ and are estimated from the standard deviation of the mean. The blue band represents the overall the ±1σ standard deviation. The reduced χ^2 for the combined data is 1.4.

experimental measurement $a_{e,exp}$ (ref. [9]) gives $\delta a_e = a_{e,exp} - a_e(\alpha_{LKB2020}) = (4.8 \pm 3.0) \times 10^{-13}$ (+1.6σ), whereas comparison with caesium recoil measurements gives $\delta' a_e = a_{e,exp} - a_e(\alpha_{Berkeley}) = (-8.8 \pm 3.6) \times 10^{-13}$ (−2.4σ). The uncertainty on δa_e is dominated by $a_{e,exp}$.

Discussion

Our measurement sets additional limits on theories beyond the standard model that lead to a contribution to a_e. Using a Bayes method[24], our result implies that for a theory with positive δa_e, we can reject $\delta a_e > 9.8 \times 10^{-13}$ with a 95% confidence level, and for a theory with negative δa_e, we can reject $\delta a_e < -3.4 \times 10^{-13}$ with a 95% confidence level.

For example, our result modifies the limits on a possible substructure within the electron. If the electron is composed of constituent particles of mass m^* bound together by some unknown attraction, its natural size should be $R = \hbar/(m^*c)$ and its magnetic moment would be modified by $\delta a_e \approx m_e/m^*$ using the simplest analysis. According to the chirally invariant model[25], our result excludes regions with $m^* < 520$ GeV/c^2 or $R > 4 \times 10^{-19}$ m with a confidence level of 95%. These are stringent limits set by low-energy experiments, although they are not yet at the limits of the Large Electron–Positron collider (the largest electron–positron collider available today)[26].

Moreover, our result sets the stage for testing whether the persistent discrepancy of 3.6σ between the experimental value[5] and the standard-model prediction of the magnetic moment of the muon[27,28] (a_μ) exists for electrons. If this discrepancy (δa_μ) is the signature of new physics, similar effects could be observable for electrons. Using naive scaling, the effects on the electron would be of the order of $(m_e/m_\mu)^2 \delta a_\mu$ (ref. [6]), where m_μ is the mass of the muon. Figure 4a summarizes the overall contributions of experiments involved in the determination of δa_e. We also include the largest theoretical contributions from the fifth order of the QED series and the hadronic term. The dominant contribution comes from the direct measurement of the electron moment anomaly, $a_{e,exp}$. For the first time, the contribution of the recoil measurement (h/m) is at the level of $(m_e/m_\mu)^2 \delta a_\mu \approx 6.5 \times 10^{-14}$, the value of δa_e deduced from the naive scaling (horizontal green bar). In the next years, improvement of one order of magnitude is expected for the accuracy of the measurement of $a_{e,exp}$ (ref. [29]); it will then be possible to probe physics beyond the standard model with comparable information from both the electron and muon.

Table 1 | Error budget on α

Source	Correction (×10^{-11})	Relative uncertainty (×10^{-11})
Gravity gradient	−0.6	0.1
Alignment of the beams	0.5	0.5
Coriolis acceleration		1.2
Frequencies of the lasers		0.3
Wave-front curvature	0.6	0.3
Wave-front distortion	3.9	1.9
Gouy phase	108.2	5.4
Residual Raman light shift	2.3	2.3
Index of refraction	0	<0.1
Internal interaction	0	<0.1
Light shift (two-photon transition)	−11.0	2.3
Second-order Zeeman effect		0.1
Phase shifts in Raman phase-lock loop	−39.8	0.6
Global systematic effects	**64.2**	**6.8**
Statistical uncertainty		**2.4**
Relative mass of ^{87}Rb[a]: 86.9091805310(60)		3.5
Relative mass of the electron[b]: 5.48579909065(16) × 10^{-4}		1.5
Rydberg constant[b]: 10,973,731.568160(21) m^{-1}		0.1
Total: α^{-1} = 137.035999206(11)		**8.1**

For each systematic effect, more discussion can be found in Methods.
[a]From ref. [13].
[b]From https://pml.nist.gov/cuu/Constants/.

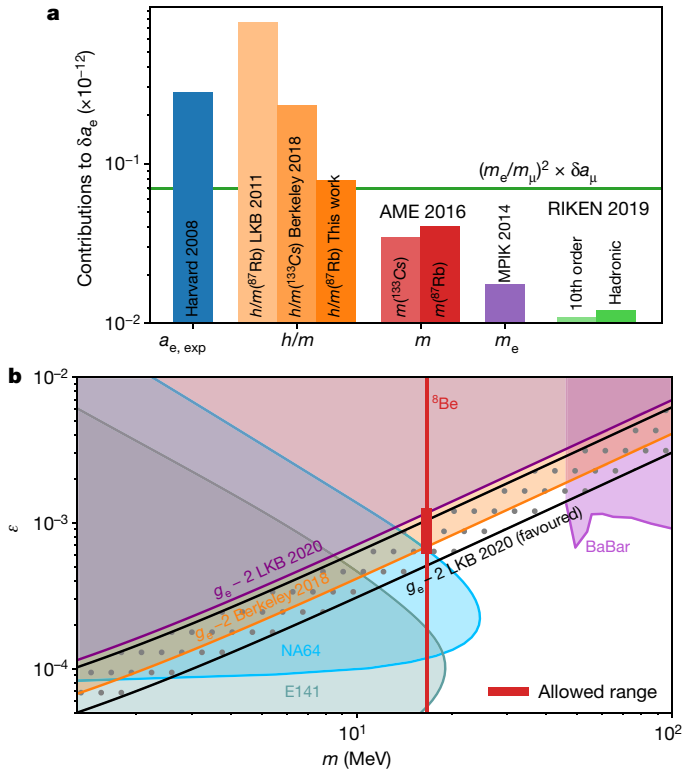


Fig. 4 | Impact on the test of the standard-model prediction of a_e and limits on hypothetical X boson. a, Summary of contributions to the relative uncertainty on δa_e. The horizontal green line corresponds to the δa_e value obtained by taking into account the muon magnetic moment discrepancy and using a naive scaling model. Previous data from ref. [9] (Harvard 2008), ref. [18] (LKB 2011), ref. [3] (Berkeley 2018), ref. [13] (Atomic Mass Evaluation, AME 2016), ref. [14] (Max-Planck-Institut für Kernphysik, MPIK 2014) and ref. [2] (RIKEN 2019). Also shown are the 10th-order and hadronic contributions in the calculation of the electron moment anomaly. **b**, Exclusion area in (ε, m_X) space for the X boson. The grey, blue and light purple regions are ruled out by the E141[31], NA64[32] and BaBar[35] experiments, respectively. A test based on the magnetic moment of the electron rules out the orange region when using the Berkeley measurement[3] and the purple region when using the present result. Disregarding the Berkeley measurement, the remaining allowed range at 16.7 MeV is depicted by the thick red line. The zone favoured by $\delta a_e > 0$, as deduced from this work, is shown by grey dots.

Finally, the anomaly reported in the angular distribution of positron–electron pairs (e^+e^-) produced in ^{8}Be nuclear transitions[4] could be explained by the emission of a hypothetical protophobic gauge boson X with a mass of 16.7 MeV followed by the decay $X \rightarrow e^+e^-$ (ref. [30]). The X boson is parameterized by a mixing strength ε with electrons and a non-zero mass m_X. Figure 4b presents the exclusion space for those parameters. At 16.7 MeV, the upper limit of ε is set by the $g_e - 2$ value of the electron and its lower limit by electron beam dump experiments (E141[31] and NA64[32] collaborations). Recently, new results from the NA64 collaboration[33] excluded ε values lower than 6.8×10^{-4}. Because vector coupling implies $\delta a_e > 0$, the result from a caesium recoil experiment imposes strong constraints on ε; combined with the NA64 result, it rejects pure vector coupling of X(16.7 MeV) at 90% confidence level. By contrast, our measurement of α gives $\delta a_e > 0$ and favours pure vector coupling with $\varepsilon = (8 \pm 3) \times 10^{-4}$, which could explain the ^{8}Be anomaly.

Online content

Any methods, additional references, Nature Research reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at https://doi.org/10.1038/s41586-020-2964-7.

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Methods

Experimental setup

The design of the science chamber is shown on Fig. 2a. A three-dimensional magneto-optical trap (MOT) is loaded by a slow atomic beam generated in a two-dimensional MOT. An optical molasses is used to further cool down atoms to a temperature of 4 μK. The temperature of the atomic cloud is measured using Doppler-sensitive Raman transitions.

After being released from the optical molasses ($t = 0$), atoms are transported to a separate chamber in which the vacuum is controlled at the level of few 10^{-11} mbar. The chamber consists of a long tube placed 50 cm above the centre of the MOT. One main difference with our previous setup[18] is that the atom interferometer is realized in this separate long tube, where the magnetic field is precisely controlled using a uniformly wound solenoid shielded by two layers of μ-metal.

Lasers for the Raman transitions are produced using second-harmonic generation from 1.56-μm lasers. These two lasers are phase-locked, and the scheme used to control the frequency difference between them during the interferometer sequence is shown in Extended Data Fig. 3a. The power used to drive Raman transitions is at maximum 70 mW per beam. The lasers are detuned with respect to the one-photon transition (Rb D2 line) by about 60 GHz.

Laser beams for the Bloch oscillations are produced from a 1.56-μm fibre laser that is split into two. Each beam seeds an optical system (μQuans) in which it passes through an acousto-optic modulator to control the laser frequency, is then amplified and passes through a periodically poled lithium niobate crystal for second-harmonic generation (about 800 mW at 780 nm). The two Bloch beams are filtered through a Rb vapour cell to reduce the resonant component of the amplified spontaneous emission of the amplifiers[36]. The total power is 400 mW for a peak intensity of 530 mW cm^{-2}. The laser is blue-detuned by 40 GHz from Rb D2 line.

The two Raman beams have linear and orthogonal polarizations. Together with one of the Bloch beams, they are transported with the same single-mode polarization-maintaining fibre at the top of the cell and pointing downwards (Extended Data Fig. 1a). A polarizing beam splitter is placed at the bottom of the vacuum cell. It transmits one of the Raman beams, which is then retro-reflected on a horizontal mirror placed on a vibration isolation table to achieve the counter-propagating configuration. The second Raman beam and the Bloch beam are rejected by the polarizing beam splitter. The inversion of the Raman effective wave vector is performed by rotating the polarization of the Raman beams by 90° before the fibre. The second Bloch beam is transported by an independent single-mode polarization-maintaining fibre at the bottom of the cell and points upwards. The waist of the beams at the output of the collimators is 4.9 mm. An apodizing filter is placed after each collimator[3].

Experimental sequence

To transport atoms in the interferometry area, we use an atomic elevator based on two Bloch oscillation pulses (acceleration/deceleration)[20]. By adjusting the parameters of the elevator (number of Bloch oscillations and delays), we can precisely choose the initial position z_0 and velocity v_0 of the cloud at the start of the interferometer $t_{\mathrm{interf.}}$. Between the two Bloch oscillations pulses of the elevator, we apply two Raman π pulses with a blow-away pulse in between. With this sequence, atoms are prepared in the magnetically insensitive state, and by controlling the parameters of the first Raman π pulse (intensity and duration) one can set the width of the vertical velocity distribution of the atomic cloud. Using a pulse duration of 189 μs, we obtain a velocity distribution with a full-width at half-maximum of 1.7 mm s^{-1}. After the preparation sequence, 500,000 atoms form the cloud.

The interferometer consists of four π/2 Raman pulses of the same duration arranged in two identical Ramsey sequences (delay T_{R}) separated by a duration T. The Bloch oscillation pulse is applied between the second and third Raman pulses (see Fig. 2c or Extended Data Fig. 1c for definitions of the pulse timing notation). To perform Bloch oscillations, we load the atoms at time $t_{\mathrm{acc.}}$ in an optical lattice by adiabatically ramping up the laser intensity for $\tau_{\mathrm{adiab.}} = 500$ μs. Then, we implement N_{B} oscillations by accelerating the lattice during time τ_{B}, which is proportional to N_{B} and in our experiment corresponds to $\tau_{\mathrm{osc}} = 12$ μs per oscillation unless otherwise specified. Finally, the lattice is adiabatically ramped down for another 500 μs.

The detection scheme (Extended Data Fig. 1b) is composed of three horizontal retro-reflected light sheets through which the atoms fall successively. The first light sheet is resonant with atoms in the state $|F = 2\rangle$ that emit fluorescence photons collected on a large-area photo-diode (F, hyperfine quantum number). A cache placed at the bottom of the light sheet blocks the retro-reflection, leading to pushing the detected atoms away from the detection system. The remaining atoms in $|F = 1\rangle$ pass through a light sheet that repumps them in $|F = 2\rangle$, and they are subsequently detected in a third light sheet similar to the first one. The relative population of atoms in each state is then obtained from the collected fluorescence signals.

Theoretical phase shift at the output of the interferometer

To maintain the resonance condition of the Raman transitions, the frequency difference ω_{R} between the lasers that drive them is carefully adjusted. In addition to the frequency difference shift $\delta\omega_{\mathrm{R}}$ between the first and third π/2 pulses, we apply during the Ramsey sequences a ramp at rate β to compensate for gravity. Thus, the effective wave vector of Raman transitions varies along the interferometer, which can induce a bias[37].

By treating this effect as a perturbation in the Lagrangian formalism[38], we obtain a modified version of equation (2):

$$\begin{aligned}\Phi = {} & T_{\mathrm{R}}\left[\varepsilon_{\mathrm{R}} 2k_{\mathrm{R}}\left(\varepsilon_{\mathrm{B}}\frac{2N_{\mathrm{B}}\hbar k_{\mathrm{B}}}{m} - gT\right) - \delta\omega_{\mathrm{R}}\right] + \phi_{\mathrm{LS}} \\ & + \frac{T_{\mathrm{R}}}{c}\left\{\beta\left[gT\left(\frac{T}{2} + T_{\mathrm{R}}\right) - v_0 T\right.\right. \\ & \left. + \varepsilon_{\mathrm{B}}\frac{2N_{\mathrm{B}}\hbar k_{\mathrm{B}}}{m}\left(t_{\mathrm{acc.}} + \tau_{\mathrm{adiab.}} + \frac{\tau_{\mathrm{B}}}{2} - T_{\mathrm{R}} - T\right)\right] \\ & \left. + 2k_{\mathrm{R}}\left(\frac{2N_{\mathrm{B}}\hbar k_{\mathrm{B}}}{m} - gT\right)\left(2v_0 - \frac{T_{\mathrm{R}} g}{2} + \varepsilon_{\mathrm{B}}\frac{2N_{\mathrm{B}}\hbar k_{\mathrm{B}}}{m} - gT\right)\right\},\end{aligned} \tag{4}$$

where k_{R} is defined as the effective wave vector when the laser frequency difference is set to address atoms at zero velocity. This formula must be used to compute h/m from the central frequency determinations of the four spectra. However, because the additional term (second and third lines in equation (4)) is independent of the direction of the Raman beams, the determination of h/m from equation (3) remains valid, provided that the value of the Raman wave vector corresponds to the one resulting from addressing atoms at zero velocity. Because we use this value, there is no correction associated to this effect.

Evaluation of uncertainty budgets

Thanks to the high sensitivity of our atom interferometer, a wide range of systematic effects was investigated and evaluated experimentally. Furthermore, we performed the measurements of h/m with various experimental parameters (N_{B}, T_{R}, τ_{B}, Raman laser intensity). The parameters are listed in Extended Data Table 1.

Given that many systematic effects depend on the position or velocity of the atoms, we implemented a Monte Carlo simulation of the experiment to calculate such effects precisely. The trajectories of the atoms during the measurement sequence were precisely controlled by means of the atomic elevator. The Monte Carlo simulation was based on the calculation of atomic trajectories using the real-time sequence of the experiment. Quantities depending on the trajectory of the atoms

(such as the contrast of Rabi oscillations or the efficiency of Bloch oscillations) were calculated and compared with experimental results to confirm the validity of the model.

Calculation of the final uncertainty

The final value of h/m was obtained from hundreds of individual measurements of h/m. For each measurement, an uncertainty was calculated. This uncertainty has several origins that may be unique to this measurement (for example, the uncertainty of the fit or the laser frequency measurement), that depend on the parameters of the measurement (for example, the light shift and the gravity gradient) or that are common to all measurements (for example, the beam parameters). The uncertainty package of Python (https://uncertainties-python-package.readthedocs.io/) was used to compute the weighted average value of h/m. The final uncertainty is a weighted quadratic sum of all the elementary sources of uncertainty. The error budget is obtained by combining those contributions according to their origin.

Monte Carlo simulation

In this simulation, each atom is described by an atomic wave packet with mean momentum $\mathbf{p}(t)$, a phase $\phi(t)$ at its mean position $\mathbf{r}(t)$, and the real amplitude $a(t)$. The momentum $\mathbf{p}(t)$ and the position $\mathbf{r}(t)$ of the wave packet evolve using classical forces that act on the atom, and the phase is calculated along this path. The sequence is split into different stages in which the accumulated phase, the evolution of the trajectory and the amplitude are computed. Three different stages are considered: free fall in the gravity field, Raman transitions and Bloch oscillations.

During free fall, the amplitude remains constant, the trajectory is given by classical physics and the phase is computed using the action along the classical trajectory.

For Raman transitions, the evolution is calculated in an accelerated frame in which the Raman frequency is constant. Analytical solutions for a finite pulse duration in the momentum representation are used[39], allowing us to compute the amplitude and the phase. The displacement is calculated from the derivative of the phase with respect to the momentum.

For Bloch oscillations, the evolution is calculated in the frame of the lattice. In this frame, the evolution is periodic and no displacement of the wave packet occurs. The phase evolution depends on three terms: (i) the phase due to the absorption and stimulated emission of N_B photons: $\phi_{ph} = N_B[\phi_{up}(x, t) - \phi_{down}(x, t)]$, where ϕ_{up} and ϕ_{down} are the phases of the two lasers of the lattice; (ii) the phase due to acceleration: $\phi_{acc} = m(g-\gamma)\tau_B/\hbar$, where γ is the acceleration of the lattice and τ_B is the total duration of the acceleration; and (iii) the phase due to the lattice, ϕ_{latt}, which is calculated from the average energy of the atom in the first band in the tight-binding limit

$$\phi_{latt} = \left[2\sqrt[4]{E_r^2 V_{up} V_{down}} + \left(\sqrt{V_{up}} - \sqrt{V_{down}}\right)^2\right]\frac{\tau_B}{\hbar}, \tag{5}$$

where E_r is the recoil energy and $V_{up/down}$ is the potential (light shift) of each individual laser of the lattice. From this energy, a classical force that acts on the atom is also calculated. The amplitude is calculated independently: the efficiency of the Bloch oscillation, which depends on both the depth of the lattice and the magnitude of the acceleration, is taken from tables computed using an independent numerical simulation[40,41].

The analytical formulas for the Raman and Bloch beam evolution are obtained assuming that the laser beams used are plane waves. Generalization to other beams is obtained by using a formula with a plane wave that locally fits the phase of the laser (amplitude, phase and phase gradient). These local parameters are obtained analytically when the simulation is performed with Gaussian beams. In the case of an arbitrary beam, numerical values are obtained using plane-wave decomposition of the solution of the Helmholtz equation. We compute the Fourier transform $\tilde{A}(k_x, k_y, z_0)$ of the wavefront at position z_0. At any position, the complex amplitude is calculated using

$$A(x, y, z) = \iint e^{i(k_x x + k_y y + \sqrt{k^2 - k_x^2 - k_y^2}\, z)} \tilde{A}(k_x, k_y, z_0)\, dk_x dk_y \tag{6}$$

and the recoil is determined using

$$k_z(x, y, z) = \iint \sqrt{k^2 - k_x^2 - k_y^2}\, e^{i(k_x x + k_y y + \sqrt{k^2 - k_x^2 - k_y^2}\, z)} \tilde{A}(k_x, k_y, z_0)\, dk_x dk_y. \tag{7}$$

The Monte Carlo simulation is performed as follows: an initial set of N wave packets (index i) is randomly calculated with a Gaussian distribution for both position and velocity. For each wave packet, and for the two paths (labelled A and B) of the interferometer, the final amplitude $a_i^{A/B}$, position $\mathbf{r}_i^{A/B}$, momentum $\mathbf{p}_i^{A/B}$ and phase $\phi_i^{A/B}$ are calculated. The phase of the interferometer is then obtained from:

$$\Phi = \frac{1}{N}\sum_{i=1}^{N} a_i^A a_i^B \left[\phi_i^A - \phi_i^B + \frac{(\mathbf{p}_i^B + \mathbf{p}_i^A)\cdot(\mathbf{r}_i^B - \mathbf{r}_i^A)}{2\hbar}\right]. \tag{8}$$

The simulation is run for each of the four spectra. The value of h/m is deduced using equation (3).

Frequency measurement

The Bloch laser and of one of the Raman lasers are locked to a Fabry–Pérot cavity. The cavity is itself locked to the two-photon transition from $5S_1/2$ ($F = 3$) to $5D_5/2$ ($F = 5$) in ^{85}Rb (ref. [42]). The frequencies of those two lasers are measured using a commercial frequency comb (MenloSystems), which is referenced by a 100-MHz signal synchronized with the French National References of Time, delivered by LNE-SYRTE via a 3-km-long optical fibre link.

Extended Data Fig. 2e presents the Allan deviation of the measurement. In the short term, we observe variations in frequency up to 4 kHz related to temperature fluctuations caused by air conditioning (timescale of about 200 s). The record of the interferometer fringes for a single h/m determination lasts approximately 300 s. For the h/m computation, we take the average of the measurements and assign to each value an uncertainty of ±4 kHz. We note that the contribution of this uncertainty on the error budget is negligible, because the laser frequency determination for two consecutive h/m measurements are independent variables, the systematic error of the frequency comb being insignificant.

The measurement was performed continuously for as long as the frequency comb laser remains locked. When the frequency measurement was not performed, but h/m measurements were taken, we estimated the frequency of the lasers using a linear interpolation and increased the uncertainty of the determination to ±8 kHz. This uncertainty is the same for a period over which the laser frequencies were not measured, that is, they did not average out. Because the frequency measurement was performed most of the time, these uncertainties contribute to a negligible amount to the error budget. The weighted mean uncertainty of the laser frequencies during this period is about 1.2 kHz per laser, leading to a contribution to the error budget of $(0 \pm 6.0) \times 10^{-12}$.

Gravity gradient

The phase shift induced by a linear gravity gradient γ is

$$\Phi_{grav.grad} = 2\varepsilon_R k_R T_R T \gamma z_m, \tag{9}$$

where z_m is

$$z_m = z_0 + \frac{v_0}{2}(T + T_R) - \frac{T\varepsilon_R \hbar k_R}{m} - g\left(\frac{T^2}{6} + \frac{TT_R}{4} + \frac{T_R^2}{6}\right) + \frac{N_B \hbar k_B}{Tm}\left[\frac{T_R^2}{12} + \frac{\tau_B^2}{12} + \left(T + \frac{T_R}{2} - t_{acc.} - \tau_{adiab.} - \frac{\tau_B}{2}\right)^2\right]. \tag{10}$$

We use the atomic elevator to position the atoms in the interferometer zone. The advantage of this method is that we can choose v_0 and z_0 so that the average position for the four spectra ($\varepsilon_R, \varepsilon_B$) is the same, thus mitigating the effect of the gravity gradient. We estimate that we can achieve a precision below 100 μm, leading to uncertainty on h/m of less than 10^{-12}. After the measurement campaign we discovered a mistake in the calculation of the trajectories for the spectra with 500 Bloch oscillations, leading to a distance of 2.31 mm between the upward and downward acceleration trajectories. Using the standard gravimeter configuration of our experiment, we measured a gravity gradient of $(2.25 \pm 0.3) \times 10^{-6}\ s^{-2}$. This leads to an overall correction on h/m of $(-1.27 \pm 0.17) \times 10^{-11}$.

Earth rotation

The Coriolis acceleration resulting from Earth's rotation induces a phase shift that depends on the individual velocity **v** of the atoms. It is given by

$$\Phi_{\text{Coriolis}} = 2T_R T \mathbf{k}_R \cdot (\mathbf{\Omega}_E \times \mathbf{v}), \tag{11}$$

where $\mathbf{\Omega}_E$ is the vector describing Earth's rotation. Its effect is expected to cancel because of the symmetry of the atomic velocity distribution. However, a residual phase shift can occur because of imperfections of the detection system[43]. Thus, we compensate for the effect of Earth's rotation by using piezo-electric transducers on the retro-reflection mirror[19,44]. We perform the calibration of the Coriolis compensation parameters by maximizing the contrast of a 700-ms-long interferometer (see Extended Data Fig. 2b).

We measured h/m with and without compensation for Earth's rotation. The result of this survey is displayed in Extended Data Fig. 2c, and shows that Earth's rotation induces a $(2.6 \pm 2.1) \times 10^{-10}$ shift of h/m. We estimate that the accuracy on the rate of the compensation ramps to ±7%, which yields an uncertainty on h/m of $(0 \pm 2.3) \times 10^{-11}$ (uncertainty obtained from the standard deviation calculated to the second order). This experimental value sets a limit on all effects related to Earth's rotation (Coriolis force and Sagnac effect).

Beam alignment

The beams are aligned automatically every 10 records of the four spectra (about 45 min). The alignment of the Bloch beams is performed by maximizing the coupling of the beam into the lower fibre. The Raman beam alignment is carried out by optimizing the retro-reflection in the upper fibre. Assuming that alignments are independent and that drifts are negligible during the 45 min between two alignments, one can calculate the standard angular deviation from the shot-to-shot alignment. We found $\sqrt{\langle\theta_R^2\rangle} \approx 3.7$ μrad between the two Raman beams and $\sqrt{\langle\theta_B^2\rangle} \approx 2.6$ μrad between the two Bloch beams. The correction on h/m given by $(\theta_B^2 + \theta_R^2)/2$ is equal to 1.0×10^{-11}. We took a conservative uncertainty of 1.0×10^{-11}.

Wave-front corrections

Using the paraxial approximation, for any scalar field $\mathcal{E}(x, y, z_0) = A(x, y, z_0)e^{i\phi(x,y,z_0)}$ at position z_0, one can calculate the local wave vector

$$k_z = k\left(1 - \frac{1}{2}\left\|\frac{\nabla_\perp \phi}{k}\right\|^2 + \frac{1}{2k^2}\frac{\Delta_\perp A}{A}\right), \tag{12}$$

where the gradient and Laplacian are taken along the x and y directions. This formula is a generalization of the analytical formula obtained for Gaussian beams.

Using a shearing interferometer, we measured the wave curvature of the beam and estimated it to be $(0.9 \pm 0.3) \times 10^{-3}\ m^{-1}$ for both beams. The term in the Laplacian (associated with the Gouy phase) was calculated from pictures taken with a camera.

Equation (12) is used in the Monte Carlo simulation to compute the local wave vector at the position of the atoms. The correction depends on the size of the cloud and therefore on the parameters used. We obtain an average value for the curvature leading to an additional correction of $(1.3 \pm 0.6) \times 10^{-11}$.

To compute the Gouy phase correction precisely, we took several pictures of each beam at different distances from the collimator corresponding to the position of the clouds in the interferometer. In the experiment, the precision on the overlap of the laser beams is less than 10% of the waist. This is not as precise as the beam alignment (less than 10^{-5}). We did not implement an automatic procedure for this less sensitive alignment.

The adjustments of the overlap between the upward and downward beams and the overlap between the atomic cloud and the laser beams were carefully made at the beginning of the measurement, and were reassessed if we observed a drop in the contrast of atomic fringes or in the number of atoms. Our method for controlling beam overlap consists of using a razor blade mounted on a micrometric translation during the alignment procedure described above. This technique, usually used to estimate the waist of the laser beam, allows us to superpose the centres of the laser beams with an accuracy relative to the beam waist better than 10%, so we took an uncertainty of ±500 μm for this overlap.

To align the laser beam overlap with the atomic cloud, we used Rabi oscillations with the two co-propagating Raman beams (downward Raman beams). The procedure consists of minimizing the pulse duration that satisfies the π condition by laterally displacing the laser beams (the fibre is placed on a micrometric translation stage). At each lateral position, the injection of the beams into the fibres was re-optimized.

Independently, simulations of the overlap of the beams between themselves and with the atomic cloud were performed using different sets of parameters. We took the standard deviation of those simulations to estimate the uncertainty of the generalized Gouy correction. We obtained an uncertainty of 2.9×10^{-11}.

In the experiment, the survival probability $P(I)$ of an atom in the Bloch acceleration process depends on the local intensity I of the laser. The effective wave vector is equal to

$$k_z^{\text{eff}} = \frac{\langle P(I(x, y, z_0))k_z(x, y, z_0)\rangle}{\langle P(I(x, y, z_0))\rangle}, \tag{13}$$

where the average is taken over the spread of the cloud. Correlation between the atomic distribution and the laser intensity leads to a systematic effect identified in ref. [21]. To precisely characterize this effect, we ran the experiment using a reduced value of I to achieve a stronger correlation between survival probability and intensity. Extended Data Fig. 4b displays the result of this experiment and the curve obtained with the Monte Carlo simulation. The parameters of the simulation were adjusted so that both the variation of the number of atoms and the variation of the relative recoil matched the experimental data. Furthermore, we added a small noise (2% at a scale of 50 μm) to the pictures recorded with the camera. We observed that for a reduction of 50% in the intensity of the laser, there is a reduction of 40% in the number of atoms and a systematic effect of about 1 ppb. Under the standard condition (90% of the intensity of the maximum intensity), the calculated correction is $(7.8 \pm 3.8) \times 10^{-11}$.

Quadratic Zeeman effect

We measured the magnetic field $B(z)$ along the vertical axis in the interferometry area using the $|F = 1, m_F = 1\rangle \rightarrow |F = 2, m_F = -1\rangle$ transition, and observed a peak-to-peak variation smaller than 50 nT. This measurement is displayed in Extended Data Fig. 2d.

The phase shift ϕ_B induced by the magnetic field can be calculated by integrating along the trajectory the variation of the Lagrangian due to the presence of the magnetic field

$$\frac{\phi_B}{2\pi} = \frac{K}{2}\int_{t_{interf.}}^{t_{interf.}+T+T_R} s(t)B(z(t))^2 dt, \quad (14)$$

where $K = 575.15\ \text{Hz G}^{-2}$, and $s(t) = 1$ if the atom is in $|F = 1, m_F = 0\rangle$ or $s(t) = -1$ if it is in $|F = 2, m_F = 0\rangle$. Using a precise computation of the atomic trajectories for each interferometer and the measured magnetic field along the interferometer area, we obtain relative corrections on h/m that are below 10^{-12}. Thus, we apply a conservative uncertainty of 10^{-12} and no correction.

One-photon light shift

The one-photon differential light shift is negative and proportional to the intensity I seen by atoms

$$\delta_{L.S.} = -\frac{I}{I_s}\frac{\Gamma^2 \nu_{HFS}}{8\Delta^2}, \quad (15)$$

where Γ is the natural linewidth of the rubidium D2 line, I_s its saturation intensity, Δ the detuning of the Raman laser with respect to the D2 line and ν_{HFS} the hyperfine frequency shift.

Along the interferometer, the intensity of each Raman pulse is increased linearly with respect to time to compensate for the expansion of the cloud and keep a constant average intensity. Typically, we increase the intensity between the first and last pulse by 10%. This number corresponds to the laser intensity ramp that reduces the velocity-dependent phase shift during Raman transitions, induced by the unbalance of the average internal intensity between Raman pulses[23]. Calibration of the slope of the intensity ramp is performed under the same experimental conditions as those of the h/m measurement to ensure the same detection efficiency.

The effects of the light shift are similar for each spectrum and therefore should cancel between the inversion of the direction of both the Raman beams and the Bloch acceleration. This compensation is not perfect for two reasons: (i) the temporal sequence being different, the size of the cloud is slightly different when the Bloch acceleration is inverted and (ii) the retro-reflected beam, which contributes twice, is different when the direction of the Raman beams is inverted. A Monte Carlo simulation is used to compute precisely the effect of the light shift and the results are shown in Extended Data Fig. 4c. These results are plotted as a function of the velocity of the cloud (detuning δ_D with respect to the resonance) and show a velocity-dependent phase shift induced by the Raman transitions[39], which has a dispersive shape. This effect has been studied in ref. [23]; it depends on the intensity and is part of the simulation.

Using the results of the Monte Carlo simulation at zero Doppler detuning, we compute the dependency of the systematic error with respect to the light shift imbalance ξ. The error then is

$$\frac{\Delta(h/m)}{h/m} \approx (\kappa_1 + \kappa_2\langle\delta_D\rangle)\xi, \quad (16)$$

where $\langle\delta_D\rangle$ is the average Doppler shift of the atomic distribution. The values of κ_1 and κ_2 are extracted from the Monte Carlo simulation for each of the eight configurations used (see Extended Data Table 1a). We estimate the light shift imbalance ξ to be 0.1 ± 0.1 and that we compensate the light shift at the velocity selection stage at ±10% of 2 kHz. As a consequence, we set $\delta_D = 0 \pm 200$ Hz. The values for each configuration are displayed in Extended Data Table 2.

Two-photon light shift

In our experiment, the two Raman lasers are transported to the vacuum cell using the same fibre (the counter-propagating configuration is then achieved by retro-reflecting one of the two lasers). Thus, co-propagating Raman transitions induce a two-photon light shift between the two hyperfine levels of

$$\delta_{2\gamma} = -\frac{\Omega^2}{2\delta_{co-pro.}}, \quad (17)$$

where Ω is the Rabi frequency of the Raman transitions, and $\delta_{co\text{-}pro.}$ is the detuning of the co-propagating transitions mainly due to the Doppler effect. Because $\delta_{co\text{-}pro.}$ changes sign with the direction of the Raman beams, this effect does not cancel with this inversion. In our configuration, this light shift is evaluated at a maximum of 5 Hz. The precise estimation of this effect is performed using a Monte Carlo simulation. The uncertainty associated to this correction comes from the knowledge of the Rabi frequency of the pulses, which we estimate to 10%. Corrections for each configuration are given in Extended Data Table 2. The correction on h/m is $(-11.5 \pm 4.7) \times 10^{-11}$.

Phase shift in Raman phase-lock loop

The setup used to control the frequency difference $\omega_R(t)$ between the two Raman lasers is presented in Extended Data Fig. 3a. It is based on a phase-lock loop (PLL) that controls precisely the phase of the beat note signal between the two lasers. The PLL is controlled with a radio-frequency generator at frequency $\omega_C + \delta\omega_{RF}(t)/8$ (where ω_C is the carrier frequency of the generator, $\delta\omega_{RF}(t) = \omega_R(t) - \omega_{HFS}$ and $\omega_{HFS} = 2\pi\nu_{HFS}$). Before the PLL, the beat note signal is mixed with a microwave generator at frequency $\omega_{HFS} - 8\omega_C$. There is a ratio of 8:1 between the frequencies of the radio-frequency generator and of the beat note.

To compensate for the Doppler shift induced by gravity, we implement a frequency ramp of rate $\beta \approx \pm 25\ \text{MHz s}^{-1}$ during the Ramsey sequences of the interferometer, where the sign of the ramp is dictated by the direction of Raman beams, ε_R. Any delay δt in the PLL system induces a shift $\beta\delta t$ in the frequency seen by atoms. A constant delay cancels over the interferometer sequence—however, because of the frequency shift between the two Ramsey sequences, this delay varies and a phase error appears.

To mitigate the bias induced by this effect, we implemented a symmetric configuration (Extended Data Fig. 3b) by changing the frequency offset ω_C with the direction of Raman beams. Specifically, we impose that $8\omega_C + (\delta\omega_{RF}(t_{interf.}) + \delta\omega_{RF}(t_{interf.} + T) + \varepsilon_R\beta T_R)/2$ is the same when the direction of Raman beams is inverted, so that the optical PLL works in similar conditions.

However, an independent measurement of the phase of the beat note signal revealed imperfect cancellation at the level. We suspect that this residual phase comes from a nonlinear response of the phase comparator of the PPL. We performed this measurement by demodulating the beat note signal and recording it on a 100-MHz-bandwidth oscilloscope (Tektronix DPO 3014). The beat note signal was lowered in frequency by mixing it with the signal of an additional microwave generator. The pulsation of this generator was set at the average $\langle\omega_R\rangle$ of the interferometry sequence so that the signal on the oscilloscope was symmetric. The time base of the oscilloscope was calibrated by analysing the signal of a 10-MHz reference.

The beat note signal is demodulated for each pulse by computing its two quadratures with a numerically computed signal. For a single interferometric sequence we obtain the contribution of the phase of the Raman lasers using the usual formula $\phi_4 - \phi_3 - \phi_2 + \phi_1$, where the subscripts refer to the pulse order. An alternative analysis based on curve fitting of the beat note signals yields identical results.

For a single determination of h/m (see Fig. 3a), we obtain for each spectrum a collection of phase shift estimations. We compute the applied phase correction by taking the average of this collection and an uncertainty deduced from its standard deviation. The correction is accounted for in the interferometric phase (equation (2)) and propagated to the value of h/m.

We confirmed that the oscilloscope did not induce a bias by reducing its bandwidth to 20 MHz, and we did not detect shifts on h/m at the level of 0.01 ppb. Moreover, we studied the relationship between the phase of the beat note and the interferometric measurement in the following way: using 600 determinations of h/m performed continuously over about 60 h, we divided the set into six quantiles sorted by the beat note phase. For each quantile, we averaged this phase and the interferometric phase. Extended Data Fig. 3c displays the relationship between these two quantities and shows that their fluctuations are identical (slope of 1). Experimentally, we also checked that this method can compensate the phase shift observed when filters with different cutoff frequencies are inserted in the PLL.

We evaluated the related correction to h/m, which depends on the interferometer parameters N_B and T_R. We note that the effect of a phase shift $\delta\Phi$ scales as $\delta\Phi/T_R$ on h/m. For $N_B = 500$, we find a correction on h/m of about -9×10^{-10} for $T_R = 10$ ms or 20 ms, which shows that the phase shift induced by frequency ramps depends on their duration. Moreover, we find a reduced effect for $N_B = 250$ at about -6×10^{-10}, which is expected because the frequency difference between the first and third pulses is ~7.5 MHz, compared to 15 MHz for $N_B = 500$. We then find a correction of $(-82.1 \pm 1.2) \times 10^{-11}$ to h/m. The uncertainty includes a conservative uncertainty of 0.01 ppb on the phase measurement with the oscilloscope.

Data availability

The datasets generated and/or analysed during the current study are available from the corresponding author on reasonable request.

Code availability

The experimental data were analysed using a self-written analysis script, which is available from the corresponding author on reasonable request.

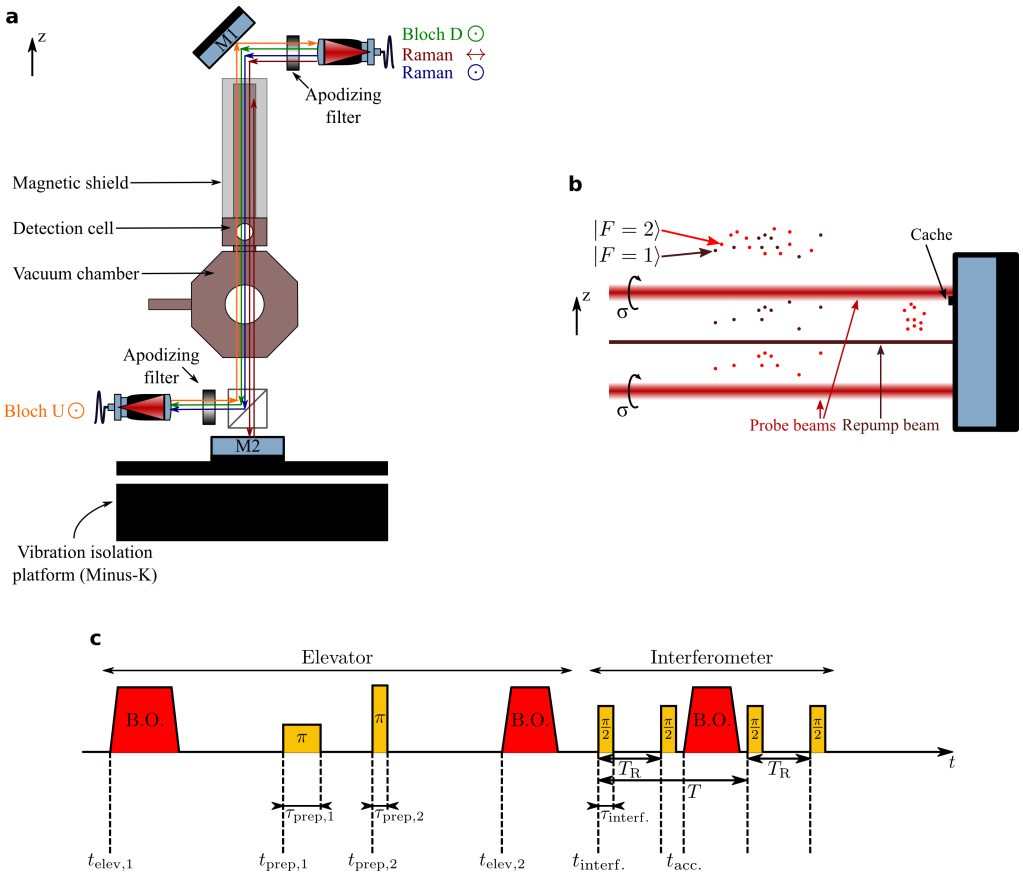
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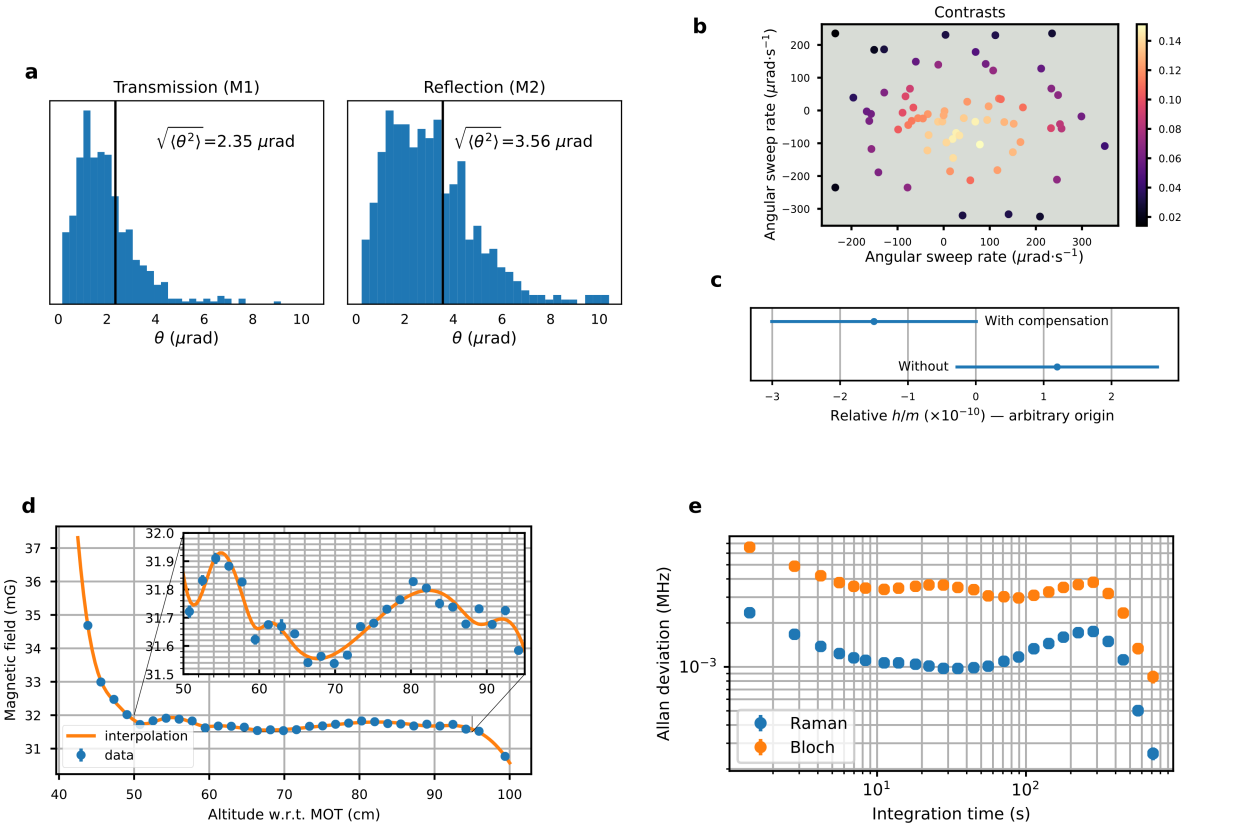
Author contributions The experiment was performed by L.M., Y.Z., P.C. and S.G.-K. The data were analysed by L.M., P.C. and S.G.-K. The main text was written by S.G.-K. and the Methods section by L.M. and P.C. All authors discussed and approved the data as well as the manuscript.

Competing interests The authors declare no competing interests.

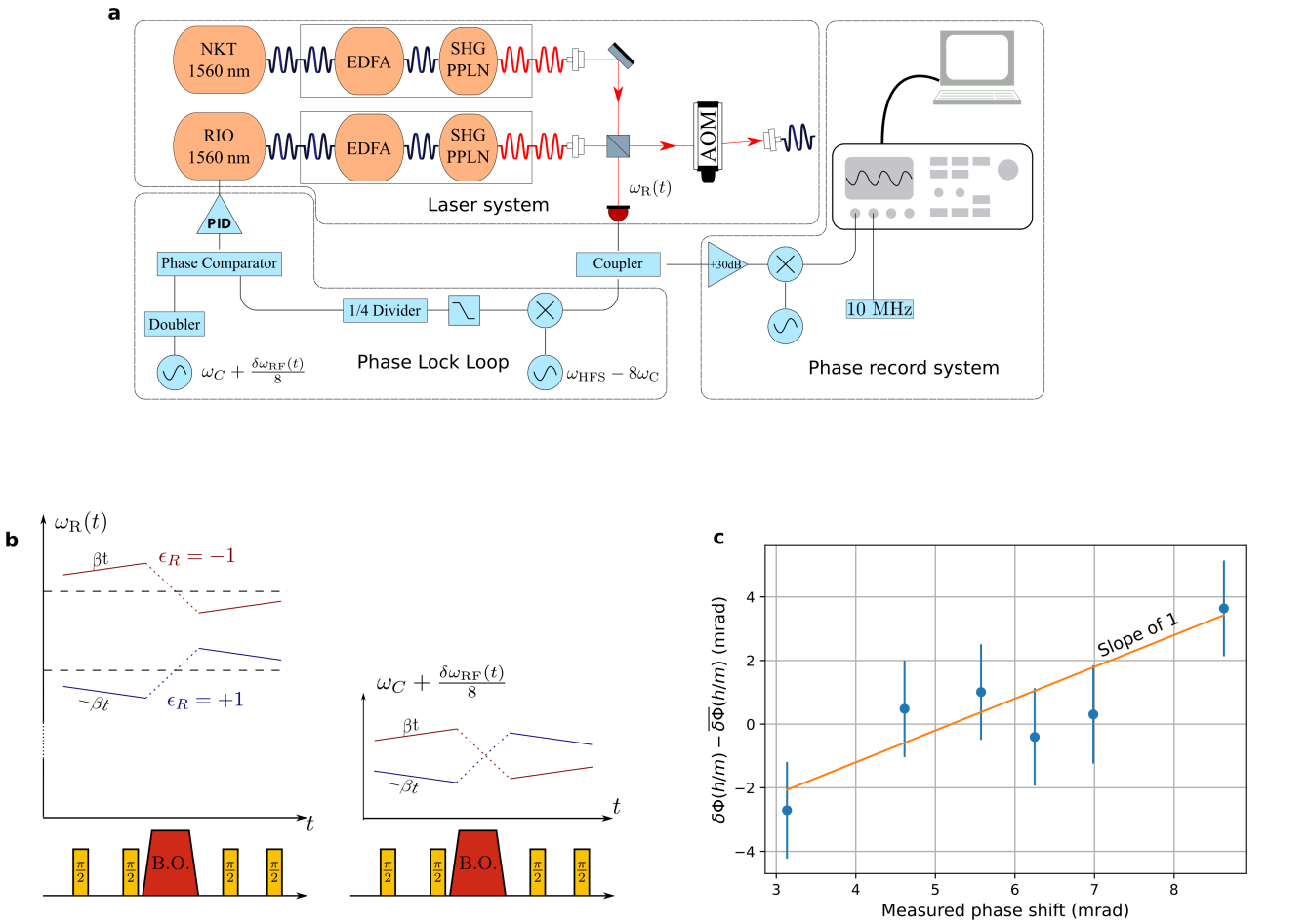
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Correspondence and requests for materials should be addressed to S.G.-K.
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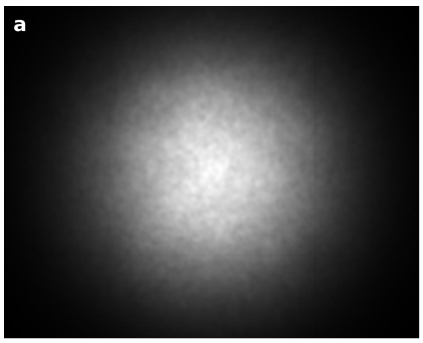
Extended Data Fig. 1 | Laser beam setup and detection. a, Vacuum cell and laser beams used for the Raman transition and Bloch oscillations. **b**, Detection setup consisting of three horizontal retro-reflected light sheets, through which the atoms fall successively. The thick red line represents the probe beams of circular polarization, which are resonant with the atoms in the state $|F=2\rangle$. The black line represents the beam that repumps atoms from $|F=1\rangle$ to $|F=2\rangle$. **c**, Light pulse sequence implemented for the measurement protocol. Shown are the temporal variables used in Methods.

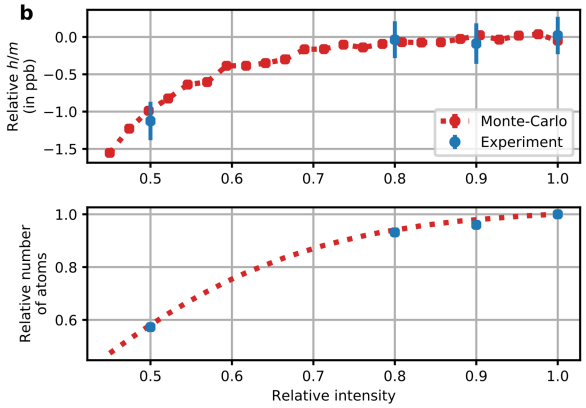


Extended Data Fig. 2 | Control of the laser beam alignment and the magnetic field. a, Distributions of the shot-to-shot variations of the auto-alignment procedure for mirrors M1 and M2 (see Extended Data Fig. 1a). **b**, Scatter plot of the contrast with respect to the sweep rate of the piezoelectric transducer of the mirror mounts (M2) for a 700-ms-long interferometer. **c**, Raw determinations of integrated h/m with and without Earth rotation compensation. Each point correspond to 400 sets of four spectra. The total interrogation time is 60 h. **d**, Blue: measured magnetic field, obtained by measuring the resonance of the magnetically sensitive $|F=1, m_F=1\rangle \rightarrow |F=2, m_F=-1\rangle$ transition. Orange: interpolation used for the modelling of the systematic effect. **e**, Allan deviation of the frequency measurement.



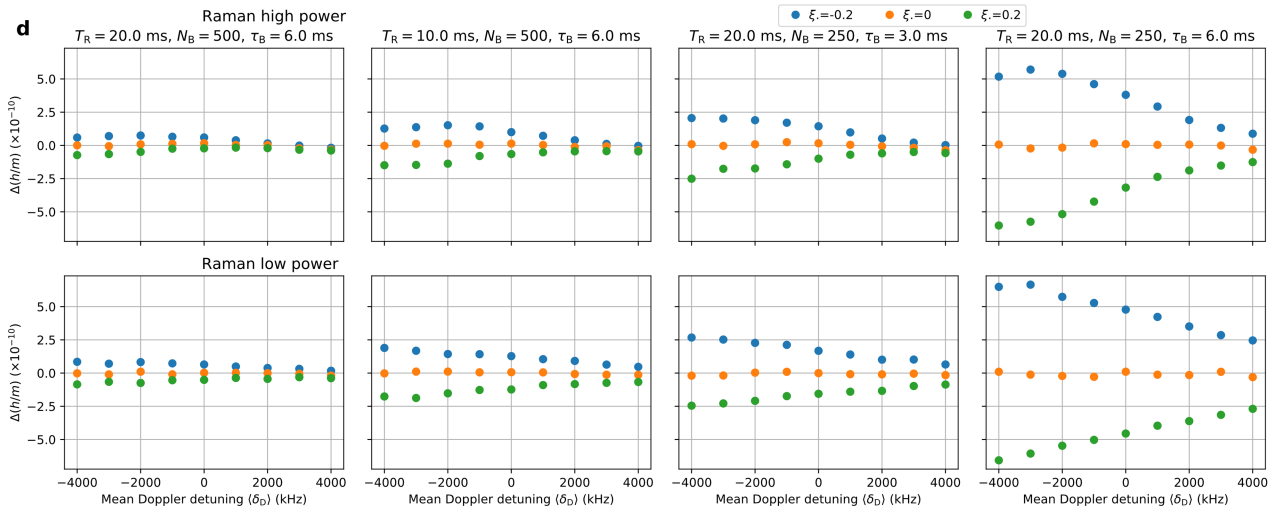
Extended Data Fig. 3 | Frequency control of Raman lasers. a, Raman phase-lock system. Top left: laser arrangement used to extract a beat note between the two lasers. Bottom left: radio-frequency chain for the phase lock. Right: setup used for the measurement of the phase between the two lasers. NKT, fibre laser from NKT photonics; RIO, diode laser from RIO lasers; EDFA, erbium-doped fiber amplifier; SHG-PPLN, second-harmonic generation using a periodic crystal; AOM, acousto-optic modulator; PID, proportional-integral-derivative controller. **b**, Frequency of the radio-frequency generator of the PLL for each Raman direction (red and blue lines). ω_C is changed with the Raman direction (right) to obtain symmetrized ramps. **c**, Average interferometric phase with respect to the average correction deduced from the phase of the beat note.





c

Raman power	Conf.	ppb
Low	A	-2.174(9)
Low	B	-2.186(6)
Low	C	-2.199(6)
Low	D	-2.123(7)
High	A	-2.14(1)
High	B	-2.155(8)
High	C	-2.181(7)
High	D	-2.092(6)



Extended Data Fig. 4 | Analysis of the effect of local fluctuations on laser intensity. a, Typical intensity profile of the laser beam. **b**, Characterization of the short-scale noise on the beam intensity. The intensity of the laser used for Bloch oscillations is reduced, leading to losses of atoms in the experiment (bottom). This induces a systematic effect on the recoil measurement (upper). To match the experimental data with the Monte Carlo simulation results, we added a small noise (2% at a scale of 50 µm) to the pictures recorded with a camera. **c**, Correction from the intensity profile calculated for each configuration. Only independent uncertainties are displayed, obtained from the Monte Carlo simulation. **d**, Results of the Monte Carlo simulation for the estimation of the effect of the one-photon light shift for different initial velocity and Raman inversion compensation (orange points: perfect compensation; blue and green points: one-photon light shift is 20% greater for one or the other Raman direction). The simulation was performed for all interferometer configurations (top: Raman high power; bottom: Raman low power) and different (T_R, N_B, τ_B) values (from left to right).

Extended Data Table 1 | Time sequence data

Configuration		Ⓐ	Ⓑ	Ⓒ	Ⓓ	
T_R		20	10	20	20	ms
N_B		500	500	250	250	BOs
τ_{osc}		12	12	12	24	µs
T		32.9				ms
$t_{\mathrm{elev},1}$		9				ms
$N_{\mathrm{elev},1}$		+650				BOs
$t_{\mathrm{prep},1}$		110				ms
$\tau_{\mathrm{prep},1}$		189				µs
$t_{\mathrm{prep},2}$		120				ms
$\tau_{\mathrm{prep},2}$		63				µs
$t_{\mathrm{elev},2}$	Spectra 1, 2	154.910				ms
	Spectra 3, 4	131.528	131.946	137.492	136.984	
$N_{\mathrm{elev},2}$	Spectra 1, 2	-1185	-1010	-1185	-1185	BOs
	Spectra 3, 4	-635	-515	-735	-735	
$t_{\mathrm{interf.}}$		175				ms
$\tau_{\mathrm{interf.}}$	Raman power: Low	80				µs
	Raman power: High	35				
$t_{\mathrm{acc.}}$		197.9				ms

Parameters used for the data collection for the different configurations and spectra. The times are indicated with respect to the release of the atomic cloud from the optical molasses. Configurations: Ⓐ T_R = 20 ms, N_B = 500, τ_B = 6 ms; Ⓑ T_R = 10 ms, N_B = 500, τ_B = 6 ms; Ⓒ T_R = 20 ms, N_B = 250, τ_B = 3 ms; Ⓓ T_R = 20 ms, N_B = 250, τ_B = 6 ms. Spectrum 1: ε_R = +1, ε_B = +1; spectrum 2: ε_R = −1, ε_B = +1; spectrum 3: ε_R = +1, ε_B = −1; spectrum 4: ε_R = −1, ε_B = −1. BO, Bloch oscillation.

Extended Data Table 2 | Light shifts

Power	Conf.	κ_1 ($\times 10^{-10}$)	κ_2 ($\times 10^{-13}$ Hz^{-1})	One-ph. corr. ($\times 10^{-11}$)	Two-ph. corr. ($\times 10^{-11}$)
Low	Ⓐ	-2.1	0.52	2.1 ± 2.1	-12.2 ± 3.7
Low	Ⓑ	-4.1	1.3	4.1 ± 4.1	-27.2 ± 5.9
Low	Ⓒ	-6.1	1.6	6.1 ± 6.1	-7.0 ± 3.0
Low	Ⓓ	-17	4.2	17 ± 17	-3.6 ± 2.7
High	Ⓐ	-2.9	0.48	2.9 ± 2.9	-39.3 ± 8.1
High	Ⓑ	-6.3	0.79	6.3 ± 6.3	-88.2 ± 17.8
High	Ⓒ	-8.1	1.3	8.1 ± 8.1	-27.4 ± 5.8
High	Ⓓ	-23	2.6	23 ± 23	-18.0 ± 4.3

Data computed with the Monte Carlo simulation for different configurations: with correction parameters of the light shift imbalance over the Raman direction (κ_1) and atomic mean velocity (κ_2); with relative corrections to h/m due to the one-photon light shift ('One-ph. corr.'); with correction on h/m for two-photon light shifts during Raman pulses ('Two-ph. corr.').
The configurations Ⓐ to Ⓓ are described in Extended Data Table 1.